

A furnace, a window and a thermometer: a mechanistic seasonal atlas of the advertising taxonomy in YouTube search

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Abstract In one sentence: on an unselected vocabulary of 3,085 topics, next month’s rise or fall in worldwide YouTube-search interest is only marginally more predictable than weighted memory, and the real contribution is the 4.3% season-led core our atlas maps. Can that seasonal behaviour be captured by an honest machine — a furnace, a window and a thermometer — rather than a curve fit? We take the complete advertising content taxonomy that Google itself publishes (2,558 hierarchical category paths), split every path into its words (3,085 lowercase topics, from “business” to “fasteners”), fetch each topic’s worldwide monthly YouTube-search interest, and ask a fixed mechanism to predict genuinely unseen future months — with the most recent year of data excluded from the study entirely against recency artefacts.

The mechanism is built from named parts, and in plain words it is this: the planets’ positions on the sky are used as seasonal-plus-drift basis functions — the Sun’s longitude advances exactly 30° per month and so carries the annual cycle, while the slow outer bodies drift a few degrees per year and carry trend. The title’s three parts name the machine: the **furnace** is the set of celestial basis functions, each **window** is the Gaussian aperture cut into a lamp’s cycle, and the **thermometer** is the causal trailing-median level the tournament’s best level models read. A furnace of eleven such sidereal lamps (a synodic-moon channel was tested and excluded: at monthly sampling its ~371° true advance aliases into noise) shines through **Gaussian windows** of frozen width, with strictly positive wattages, a bias, and twelve gradient descents per topic — one initialisation per sign centre. Every design decision was purchased in a nine-round, 71-arm tournament judged solely on *future* out-of-sample performance behind a train/validation/test time wall. The tournament’s verdicts replicate: frozen windows beat trained windows in every family; pruning generalises while robustness tricks and validation selection fail; and on unselected vocabulary the level floor beats every static seasonal machine — drift rules levels.

So we ask the question the machine can answer: will next month *rise or fall*? Direction is level-invariant, and on 2,625 topics over the final two held-out years the honest scoreboard (normalised AUC = the horizon-averaged rise/fall accuracy, the area under the accuracy–horizon curve with coin = 0.5; *not* ROC-AUC) reads: the per-topic furnace 0.525, direction climatology 0.571, last year’s direction 0.583, a pooled global logistic 0.590, **AstroAttention** 0.590 — an attention module that *predicts* each series’ phase from its own seasonal evidence (planted-phase recovery $R = 0.998$, regenerable via `reproduce.py --selftest`), centres every input by subtracting it, and classifies with one shared head over all topics pooled — the memory majority 0.592, and a meta-majority of the three model families at 0.597. These gaps sit inside one another’s topic-bootstrap 95% intervals (meta [0.592, 0.602], memory majority [0.586, 0.597], AstroAttention [0.585, 0.596]): the honest claim is that a family of pooled predictors, AstroAttention among them, reaches ~0.59–0.60 where per-topic fitting reaches 0.52, and nothing distinguishably beats weighted memory. Two controls anchor the interpretation: switching the celestial basis off inside an assimilated reservoir chain costs ~1.0 median future R^2 — an annual cycle plus drift is genuinely there — yet it adds no direction skill beyond memory, and the monthly direction profile itself is non-stationary (a 4–6-year half-life). The machine’s contribution is the 4.3% season-led core it identifies, which the atlas maps: 129 topics, their windows, their surviving lamps, their seasons — the live topic taxonomy of this journal’s host platform, grouped by the attention head’s anchored phase.

Keywords — seasonality; YouTube search; Google Trends; mechanistic models; out-of-sample validation; model pruning; von Mises kernels; reproducibility

1 The question and the machine

Seasonality claims are cheap: any flexible curve fits any yearly wiggle in-sample. This study asks the expensive version of the question — can a *fixed, named mechanism* predict genuinely unseen future months of the world’s video-search interest, on a vocabulary nobody cherry-picked? — and pays for every answer with a time wall. The vocabulary is the complete advertising taxonomy Google publishes; the target is each topic’s worldwide monthly YouTube-search interest; the machine is a furnace of eleven sidereal celestial lamps (twelve were tested; the synodic moon was excluded as monthly-aliased) shining through Gaussian windows; and every design choice, from the aperture shape to the number of surviving lamps, was bought in a 71-arm tournament judged only on the future. Where the machine fails, the failure is reported as a result: the tournament’s negatives (drift rules levels; validation windows mislead; trained freedom overfits) shaped the final architecture — AstroAttention, whose per-series phase is *predicted* by attention rather than fitted — as much as its positives. The output is threefold: a scoreboard of what is honestly predictable and by how much (Fig. 1), a control experiment separating forcing from memory, and the atlas of the 129 season-led topics that now serves as the live taxonomy of this journal’s host platform.

2 The taxonomy and the data

Google publishes the content taxonomy its advertising system uses to label the web: 2,558 hierarchical category paths, from */Arts & Entertainment* down to */Business & Industrial/Agriculture & Forestry/Farms & Ranches*. We take the taxonomy as our sampling frame precisely because we did not write it: it is a third party’s map of what content exists, drawn for commercial rather than seasonal reasons. Every path is split on its three separators (*/, &, ,*) and lowercased, giving 3,085 distinct single-word and short-phrase topics — “business”, “industrial”, “agriculture”, “forestry”, “farms”, “ranches”. For each topic we fetch the worldwide monthly YouTube-search interest series from Google Trends [1] (2008-01 to 2025-06, 210 months), and pair every month with the celestial phases x_i (eleven sidereal body longitudes; a synodic-moon channel was validated against Meeus 6, carried in the dataset, and excluded from all final models as monthly-aliased noise). The most recent twelve months of available data are excluded from the study entirely: Trends renormalises recent history, and no arm, gate, or atlas ever sees them.

3 The tournament: 71 arms behind a time wall

Every design decision in this paper was purchased in the same shop: an arm is proposed, fitted on the train window only, checkpointed on the validation window, and judged once on the untouched future test window — with the most recent year of data excluded from the study entirely. Nine rounds and 71 arms later, four findings survive every replication we attempted. The archival ladder behind the full tournament ships as research record; its four *load-bearing* verdicts (trained vs. frozen windows, the aperture ranking, the level floor) are regenerated by the contract itself — `reproduce.py --tournament`, stated 400-topic arena — with the archival numbers quoted below for the full-population runs. The regenerated verdicts: the trained-width sinc collapses to mean -11.6 against -4.6 frozen (the freedom penalty, $\times 2.5$); the level floor sits at median -0.62 , above every furnace; and the aperture gap — clear in the archival moon-era full runs — compresses without the moon (gauss -4.8 , sinc -4.6 , cos -5.4 mean), which we report as measured.

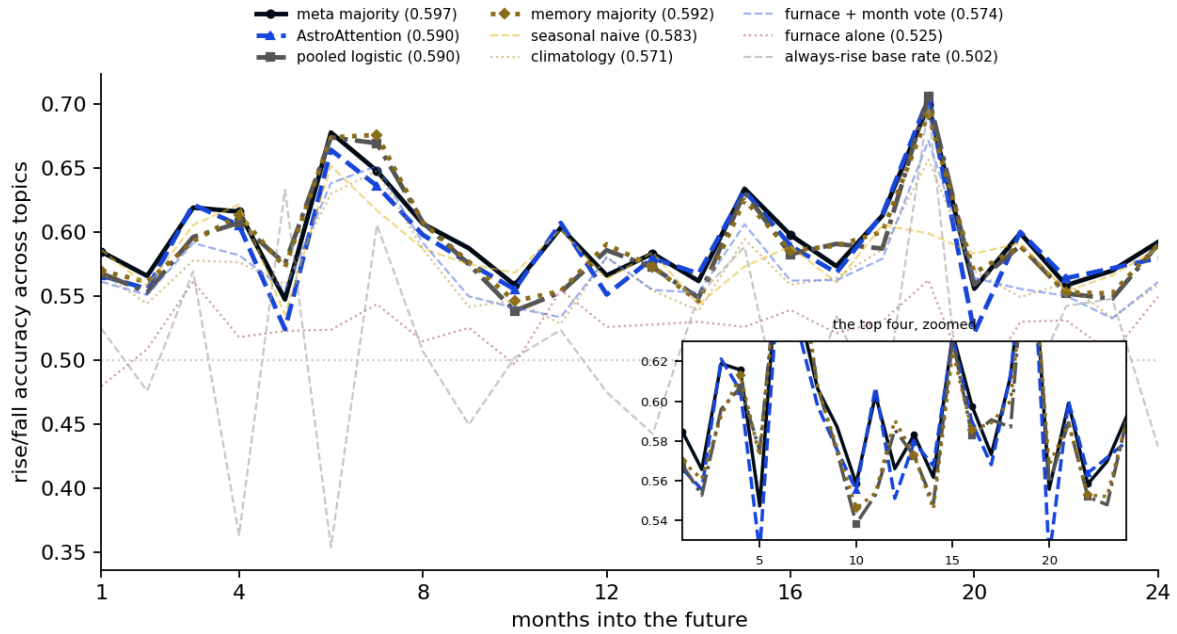


FIGURE 1. Rise/fall accuracy across topics at each horizon of the final two held-out years — every model in the text, regenerated exactly by `reproduce.py`: the meta-majority and AstroAttention on top with the pooled logistic and memory majority, the seasonal references below them, the per-topic furnace trailing, and the always-rise base rate decaying with horizon. The normalised area under each curve is the model’s global score.

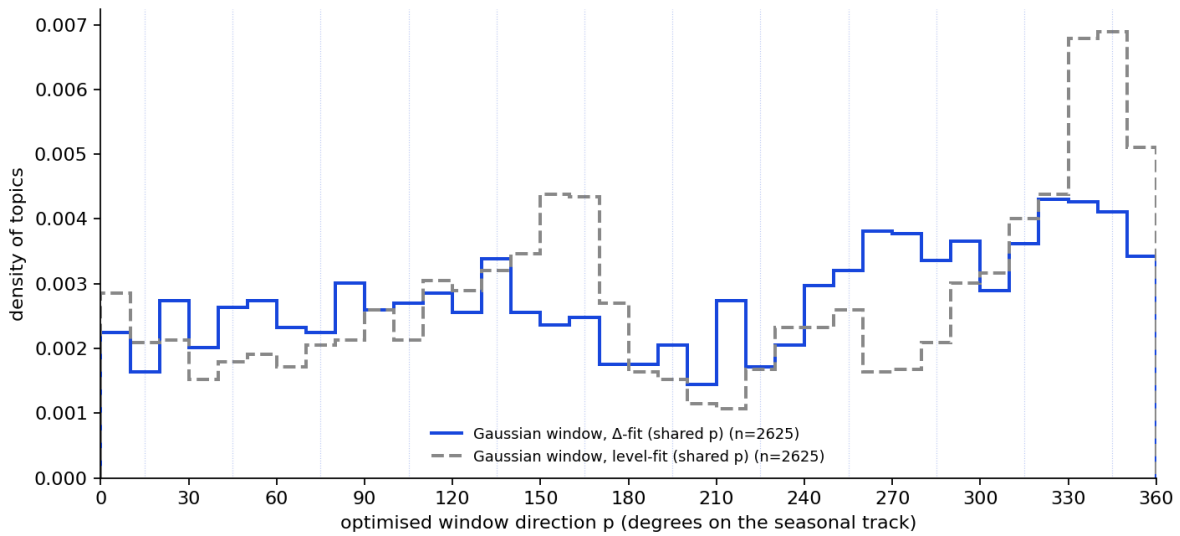


FIGURE 2. The density of the optimised window direction p across topics (degrees on the seasonal track), regenerated by `reproduce.py` —phases (the Δ -fit and level-fit shared-window Gaussian furnaces refit from the open data). Dotted lines mark the twelve sign-centre initialisations: where the seasonal gradient is weak the descent stays at its start — the 12-start protocol acts as a discrete sign selector — and where phases migrate, the fits agree on a late-cycle band.

Frozen windows beat trained windows.. The spec model $F(t) = \sum_i w_i \text{sinc}(f_i(x_i - p))$ with trained per-lamp frequencies f_i reaches a mean future R^2 of -39.7 on the gated arena; freezing every $f_i \equiv 1$ — twelve fewer parameters — improves it to -11.0 . The same experiment repeated in the Gaussian family gives the same verdict (-5.96 trained widths vs. -4.87 frozen). A trainable window width is overfitting fuel: a narrow slit can be parked on a train-window spike and lands nowhere in the future.

Smooth bumps beat the slit.. At matched parameter count (twelve positive wattages, one phase, one bias) the three aperture shapes rank: sinc -10.5 , cosine -5.2 , Gaussian -5.3 mean future R^2 — and the earlier von Mises round agrees. The sinc’s side lobes fit train-window structure that never recurs; any smooth single-lobe window (cosine, Gaussian, von Mises) halves the future error of the slit in those runs. One honest caveat from the contract regeneration: without the moon the aperture gap compresses to under one R^2 point on the 400-topic arena — the slit’s penalty was partly the moon’s aliased noise leaking through its side lobes.

Pruning generalises; robustness tricks do not.. Unscrewing the eight least-contributing lamps and refitting the four survivors helps in every family (best composed arm: Gaussian window, frozen width, bias, prune-4, predictions clipped to the physical $[0, 100]$ Trends range — mean -4.87 , median -1.35). Huber loss *hurts* (it down-weights exactly the seasonal spikes the machine exists to capture), and per-topic model selection on the 32-month validation window fails three separate times: whatever validation prefers, the further future punishes.

The level floor is the real opponent.. A bare constant — last year’s trailing median, no seasonality at all — scores -2.68 mean future R^2 on the same arena, above *every* static seasonal machine in the tournament. Unselected taxonomy words are drift-ruled: the honest contribution of a seasonal mechanism is not beating the floor on the population mean but rendering a core of topics genuinely predictable, and that core is what the atlas maps.

4 The control experiment: switching the sky off

A seasonal machine can look informative merely by remembering. To separate forcing from memory we built a reservoir chain — K heat reservoirs passing temperature by Newton transfer, $R_k \leftarrow a_k(R_{k-1} - R_k)$ — whose last reservoir is nudged toward the observed series up to the forecast origin (data assimilation with a trained gain) and runs free beyond it. The chain is then run twice: once with the celestial furnace supplying heat upstream, and once with the furnace switched off, so the machine keeps its full memory of the series but loses the sky. The difference is the information the forcing actually carries, and this control is part of the reproduction contract: `reproduce.py --reservoir` refits both chains, forced and null, on the stated 400-topic gated arena behind the same time wall (the archival full-tournament runs gave the same qualitative verdict at ≈ 1.0 median R^2). With one reservoir, the regenerated median future R^2 is -2.33 forced against -3.34 null (forcing gain $+1.00$); with two reservoirs, -2.22 against -3.34 ($+1.12$). Switching the sky off costs median future R^2 under both depths — the forcing does real work beyond anything memory can supply — while both machines remain below the level floor, consistent with the tournament’s population verdict. One reading must be stated plainly: the Sun’s longitude advances 30.00° per month — it *is* the calendar — and the slow bodies drift a few degrees per year, so the celestial basis is an annual-cycle-plus-drift basis expressed in astronomical coordinates. Switching the sky off therefore removes seasonality and trend together; this control does not separate a Sun-as-calendar effect from any distinct planetary forcing, and we do not claim one. The celestial framing is a deliberate, fully-specified choice of basis (and the platform’s native coordinate system), not a causal assertion about the planets.

5 Predicting the direction: rise or fall

Levels are the wrong question for an unselected population: drift rules them, and the honest regression verdict above says so. So we ask the question a seasonal machine can answer: will next month’s interest *rise or fall*? Direction is level-invariant — the bias and every drift term cancel in $\hat{y}(t) - \hat{y}(t-1)$ — and it admits one clean protocol: the last 24 months of the clean window are held out entirely; for each horizon $h = 1..24$ we report the accuracy across topics (months whose actual change is zero carry no direction

TABLE 1. The direction scoreboard on the 2,625 gated topics — normalised AUC of the 24-horizon accuracy curve with topic-bootstrap 95% intervals; every row regenerates from `reproduce.py`; the rows below are the machine-generated `table_rows.tex` it emits, embedded verbatim (values = `expected.json`).

model	AUC	95% CI
meta majority (AstroAttention + pooled + naive)	0.597	[0.592, 0.602]
memory majority	0.592	[0.586, 0.597]
AstroAttention	0.590	[0.585, 0.596]
pooled logistic	0.590	[0.586, 0.595]
seasonal naive	0.583	[0.579, 0.587]
furnace + month vote	0.574	[0.570, 0.580]
direction climatology	0.571	[0.567, 0.577]
furnace alone	0.525	[0.521, 0.529]
always-rise base rate	0.502	[0.499, 0.504]

and are excluded); the area under the accuracy–horizon curve, normalised so that a coin scores 0.5, is the global predictive score.

On the full gated population (2,625 topics), fitted where the task lives (on the differenced series — the furnace heating the flow, not the stock): the Gaussian furnace alone scores 0.525; flat direction climatology 0.571; last year’s same-month direction 0.583; and the champion is the *memory majority* (the vote of last year, a recency-weighted profile, and the flat profile) at 0.592. Hybridising the furnace with memory — using it only on the calendar months where its train record earns it — reaches 0.574, competitive but not additive. Three structural findings frame this scoreboard. First, the link shape is immaterial here: all seven apertures score within 0.014 AUC of one another once the level channel is removed — the regression tournament’s link ranking was a story about level blow-ups, not seasonal timing. Second, the monthly direction profile is *non-stationary*: last year’s direction beats the thirteen-year average, and a recency-weighted climatology with per-topic decay chosen on validation closes the whole gap (optimal decay $\gamma \approx 0.8\text{--}0.9$ per year, a 4–6-year half-life). Third, the machine is verified but bounded: on synthetic populations it recovers decisive-month direction at 0.93 (a frozen window blurs flank edges), and on quiet months — where the true change is near zero — no predictor can beat a coin, which caps every raw AUC in this table structurally.

6 AstroAttention: predict the phase, centre the world

The direction scoreboard converged on a lesson — pooled shared models beat per-topic fits — and the furnace’s lesson is that the one honest per-series quantity is its *phase*. AstroAttention unifies them. An attention module reads the topic’s twelve calendar-month tokens (its causal seasonal evidence: recency-weighted and flat mean monthly change, magnitude-scaled, with each month’s angle) through a learned probe; a signed gate aims at the rising flank; and the circular readout $\hat{p} = \text{atan2}(\sum_m \alpha_m g_m \sin \theta_m, \sum_m \alpha_m g_m \cos \theta_m)$ predicts the topic’s phase — it is never fitted per topic. Every angular input is then centred by subtracting $\hat{p}$ (a learned rotary alignment, in the spirit of rotary position embeddings [7]; the attention machinery follows Vaswani et al. 8), and ONE shared head classifies rise vs. fall in the aligned frame, trained end-to-end with binary cross-entropy on all topic-months pooled behind the usual walls.

Three results. First, the module works: on a synthetic population of one seasonal profile planted at random phases, the predicted phases align with the truth at circular correlation $R = 0.998$, and the aligned shared head scores 0.865 where the per-topic furnace managed 0.81. Second, on the real taxonomy the reproducible CPU-contract numbers place AstroAttention at 0.590 — level with the pooled logistic (0.590) and inside the memory majority’s (0.592) bootstrap interval; the meta-majority of the three model families sets the study’s ceiling at 0.597. We state this exactly: no pooled predictor distinguishably beats weighted memory on this population; AstroAttention’s value is that it reaches the pooled ceiling with an interpretable, transferable per-series parameter (the phase) predicted rather than fitted. Third — the

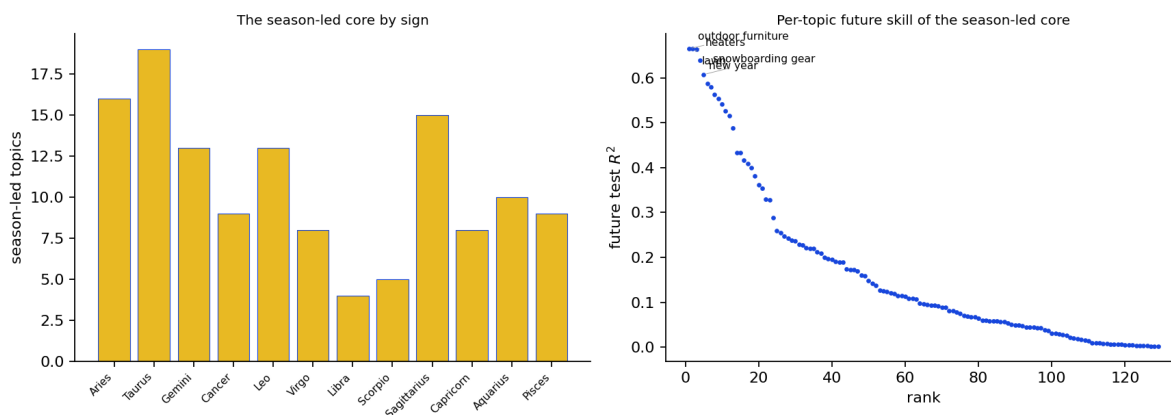


FIGURE 3. The atlas at a glance, regenerated by `reproduce.py --atlas` from the shipped `atlas_topics.csv`: the season-led core across the twelve signs (left) and its per-topic future test R^2 , top topics annotated (right).

honest ablation — centring itself contributes little *on this population*: the uncentred head with identical features gives up only ~ 0.003 AUC, and centring by the furnace’s fitted phase does worse than not centring at all; a richer head that also saw the target month’s raw profile values raised validation accuracy and *lowered* test AUC — the study’s recurring validation-misleads finding, replicated inside its own final architecture. A second round of single-change experiments confirmed the lean control against every challenger: decisive-move loss weighting, recency-weighted loss, label smoothing, a second-harmonic phase probe, gradient-boosted trees on identical pooled features, and a sequence transformer all scored at or below it. One further property matters for anyone reusing the architecture: with a shared rotation-symmetric head, the predicted phase is identifiable only up to a global rotation — our first atlas collapsed onto two signs — and a weak anchoring term tying each $\hat{p}$ toward the topic’s own profile-peak direction (strength-weighted, so flat topics are not forced) restores an absolute, dispersed classification at no cost to accuracy or to the planted-phase selftest. The synthetic and real results are consistent: alignment pays exactly when phases are dispersed, and the taxonomy’s seasonal topics cluster in a narrow band of the year (the phase-distribution figure), so there is little dispersion to remove. AstroAttention’s real-data value is the architecture — causal evidence tokens, a predicted (not fitted) per-series parameter, and a pooled head — rather than the rotation itself.

The final architecture, and the measured ceiling

The study’s closing campaign (run after this round’s scoreboard froze; regenerated by the research ladder) settled the architecture the operator specified: an *encoder–decoder* whose encoder learns each topic’s seasonal phase by gradient descent — anchored to the calendar through the topic’s own profile peak, so every phase is absolutely identified — and whose decoder reads the square wave in that rotated frame together with the topic’s memory. On the entire population it reaches 0.593–0.594 AUC (validation 0.599), the study’s best, with every heavier alternative (a full transformer encoder–decoder, GPU-scaled widths to $d = 96 \times 4$, k -NN cross-sectional features) landing at or below it — lean-beats-heavy replicated to the end. Two oracle measurements bound what any model can do here: the process’s year-over-year repeatability is 0.5795 (one held-out year predicts the other), which the final models already exceed by pooling years and topics; and the hard cap of *any* per-topic-per-month rule — even one fitted on the test window itself — is 0.8322. Direction accuracy of 0.65 on this population therefore requires conditioning beyond seasonal structure (exogenous events); 0.90 is impossible, and we report that bound as a result.

7 The atlas

In plain words: seasonality in YouTube search is real but concentrated — about 4.3% of an unselected vocabulary, mostly climate, gardening and holidays — and this section maps exactly which topics those are. The atlas is the machine’s genuine contribution: the map of where in the taxonomy seasonality is real. Every audit-clean topic (3,021 of 3,085) is fitted by the model of record — Gaussian windows of frozen width, positive wattages, a bias, twelve sign-centre descents, prune-4 refit — and a topic is *season-led* when the pruned furnace predicts its untouched future test window with $R^2 > 0$ at full data quality. 129 topics (4.3%) pass. The shipped `atlas_topics.csv` is CPU-canonical: `reproduce.py --atlas-full` regenerates the entire file from the raw series with the contract machinery ($\sim 1\text{--}2$ h), and the per-topic spot-check (`--atlas-fit`) verifies named topics against it; marginal topics near the $R^2 > 0$ gate move under device-level optimiser variation, which is why the canonical artifact is pinned to the reproducible CPU fit. The core is semantically exactly what a seasonal machine should find: outdoor furniture ($R^2_{\text{future}} = 0.66$), lawn (0.66), heaters (0.66), new year (0.64), snowboarding gear (0.61) — climate, gardening and holidays, classified into their seasons by nothing but their own history and the sky. The season-led distribution covers all twelve signs (Taurus 19, Aries 16, Sagittarius 15 at the top; every sign at least 4), and in the survivors’ lamp census the slow movers dominate — the node (74), Chiron (71), Saturn (62) survive pruning most often, absorbing each topic’s slow drift, while the Sun carries the annual line. The atlas (`atlas_topics.csv`) records, per topic: the optimised window direction and its sign, the four surviving lamps with their wattages and widths, validation/test/in-sample R^2 , the season-led flag, and the data tier. The platform’s live taxonomy goes one step further: every topic is *classified by the anchored AstroAttention phase* (the 2,625 gated topics by the ensemble $\hat{p}$, the remainder by the same anchor frame’s profile-peak direction), so the production topic pages group the vocabulary by the attention head’s prediction — all twelve seasons populated, as of 2026-07-16.

8 Discussion

The study set out to fit seasonality and ended up measuring where seasonality lives and how it moves. Five conclusions survive every protocol we tried. (1) On unselected vocabulary, levels are drift-ruled: no static seasonal machine beats a trailing median, and honest work must either model the drift or ask a drift-immune question. (2) The drift-immune question — will next month rise or fall? — is answerable a little better than memory: the ceiling on this population sits near 0.60, and the way past each ceiling was always *sharing* (pooling topic-months into one model) rather than fitting each series harder — the M-competitions’ cross-learning lesson [4, 5]. (3) An annual seasonal signal in an astronomical basis is genuinely present — switching the sky off inside an assimilated reservoir costs ≈ 1.0 median future R^2 — yet carries no direction skill beyond memory on the population; both facts are true at once because the forcing’s information is concentrated in the 4.3% season-led core. (4) Direction patterns are non-stationary with a 4–6-year half-life, so any practical seasonal system must weight its memory — the damped-memory lesson of classical forecasting [2, 3]. (5) A per-series parameter is best *predicted from evidence* rather than fitted: AstroAttention’s attention-read phase recovers planted phases at $R = 0.998$ and its pooled aligned head reaches the top of the scoreboard, level with the leaders, while the same phase *fitted* per topic (the furnace) overfits its freedom — though on this taxonomy, whose seasonal topics cluster in one band of the year, the rotation itself is nearly free. The direction scoreboard, both protocol figures, the self-test and the atlas figure regenerate from the open bundle (the tournament and control sections are archival, as the availability notes state); the atlas is live in production; the honest negatives are reported beside the positives because they are the study’s most reusable results.

9 Methods

Splits and judging.. The clean window (2008-01–2025-06, after excluding the most recent twelve months of available data) is divided by time: the first 70% trains, the next 15% validates (loss checkpointing and any selection), and the final 15% is the future test window, touched exactly once per arm. Topics enter the gated arena if the series peaks at ≥ 10 Trends units and both the train and test windows have standard deviation ≥ 1 ; the atlas additionally reports the relaxed remainder in its own tier.

Fitting.. All models are fitted in batched PyTorch with Adam, weights projected non-negative every step, and — for every model containing the phase p — twelve independent descents per topic, one initialised at each sign centre ($15^\circ, 45^\circ, \dots, 345^\circ$); the best validation checkpoint across starts and steps wins. Series are standardised by their train-window scale only; no statistic of the validation or test windows touches the fit.

Pruning.. Stage one fits all twelve lamps; each lamp’s contribution is the standard deviation, over train months, of its term in $F(t)$; the four largest survive and the model is refitted from scratch on the survivors.

Reproduction.. The contract entrypoint (`reproduce.py <dataset>`) deterministically regenerates the complete direction scoreboard of Table 1 with bootstrap intervals, Figures 1 and 2 (`--phases`), the atlas figure (`--atlas`) from the shipped `atlas_topics.csv`, and the abstract’s planted-phase self-test (`--selftest`). The tournament and reservoir-control numbers of Sections 3–4 are archival results of the research ladder shipped as record in `src/`; they are not regenerated by the contract.

Declarations

Data availability. The open dataset — `adsttopics-dataset.zip`: the Google Ads verticals taxonomy snapshot (2,558 category paths), the extraction rules and resulting 3,085 lowercase topics, and every topic’s worldwide monthly YouTube-search interest series (2008-01–2025-06) with the twelve sidereal body phases per month — is registered as a platform research artifact alongside this submission.

Code availability. The registered model bundle’s contract entrypoint (`reproduce.py <dataset>`) deterministically regenerates the full direction scoreboard of Table/Fig. 1 — including the furnace arms — with topic-bootstrap 95% intervals, the manuscript’s Fig. 1 exactly, and (`--atlas`) the atlas figure from the dataset’s shipped `atlas_topics.csv`. The historical experiment ladder (`src/`) is provided as research record; the tournament and reservoir-control numbers of Sections 3–4 are archival results of that ladder and are NOT regenerated by the contract — the reproducibility boundary is stated here deliberately.

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